Google App Engine Java 17 Setup Guide

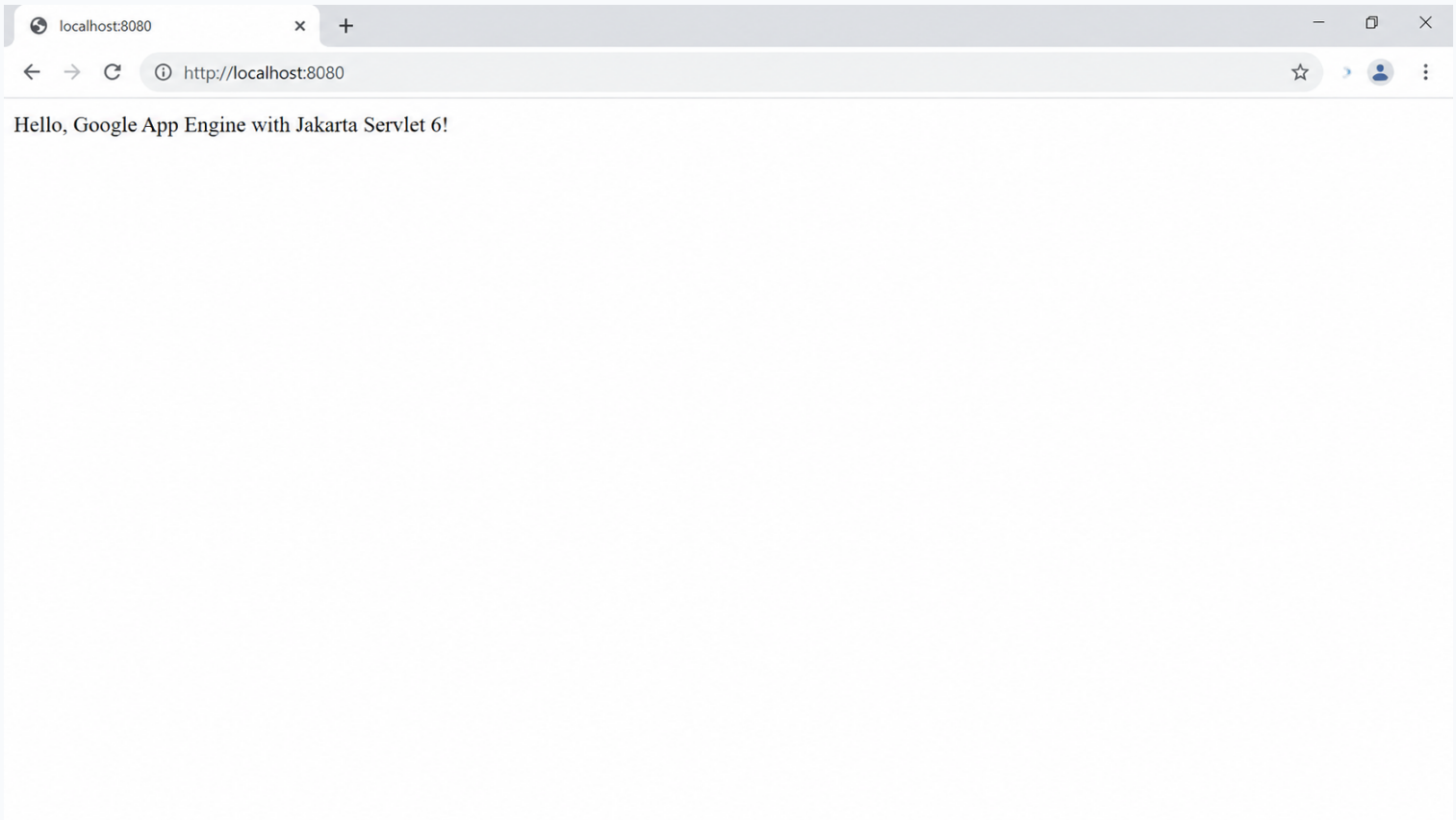
Complete step-by-step instructions for Jakarta Servlet 6.0, Jetty 12 local development, and GCP deployment.

Java 17 | Jakarta EE 10 | Servlet 6.0 | Jetty 12.0.8 | App Engine Standard

1. Maven Project Configuration (pom.xml)

Place this corrected and structured Maven configuration in `pom.xml` in your project root. It includes the required Maven compiler settings, WAR packaging plugin, App Engine deployment plugin, and the Jetty 12 EE10 plugin for local execution.

OUTPUT :



```xml
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/
maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com.example</groupId>
    <artifactId>gae-java-hello</artifactId>
    <version>1.0-SNAPSHOT</version>
    <packaging>war</packaging>

    <name>gae-java-hello Maven Webapp</name>
    <url>http://www.example.com</url>

    <properties>
        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
        <maven.compiler.source>17</maven.compiler.source>
        <maven.compiler.target>17</maven.compiler.target>
    </properties>

    <dependencies>
        <dependency>
            <groupId>jakarta.servlet</groupId>
            <artifactId>jakarta.servlet-api</artifactId>
            <version>6.0.0</version>
            <scope>provided</scope>
        </dependency>
    </dependencies>

    <build>
        <finalName>gae-java-hello</finalName>

        <plugins>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-compiler-plugin</artifactId>
                <version>3.11.0</version>
                <configuration>
                    <source>17</source>
                    <target>17</target>
                </configuration>
            </plugin>

            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-war-plugin</artifactId>
                <version>3.4.0</version>
            </plugin>

            <!-- Google App Engine Plugin -->
            <plugin>
                <groupId>com.google.cloud.tools</groupId>
                <artifactId>appengine-maven-plugin</artifactId>
                <version>2.8.0</version>
            </plugin>

            <!-- Jetty 12 Plugin for EE10 / Servlet 6.0 -->
            <plugin>
                <groupId>org.eclipse.jetty.ee10</groupId>
                <artifactId>jetty-ee10-maven-plugin</artifactId>
```

```
                <version>12.0.8</version>
            </plugin>
        </plugins>
    </build>
</project>
```

2. App Engine Configuration (appengine-web.xml)

Create file at path: `src/main/webapp/WEB-INF/appengine-web.xml`

```
<?xml version="1.0" encoding="utf-8"?>
<appengine-web-app xmlns="http://appengine.google.com/ns/1.0">
  <runtime>java17</runtime>
  <system-properties>
    <property name="java.util.logging.config.file" value="WEB-INF/logging.properties"/>
  </system-properties>
</appengine-web-app>
```

3. Web Deployment Descriptor (web.xml)

Create file at path: `src/main/webapp/WEB-INF/web.xml` using Jakarta EE 10 / Servlet 6.0 schema.

```
<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns="https://jakarta.ee/xml/ns/jakartaee"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="https://jakarta.ee/xml/ns/jakartaee https://jakarta.ee/xml/ns/jakartaee/
web-app_6_0.xsd"
         version="6.0">

    <display-name>gae-java-hello</display-name>

    <servlet>
        <servlet-name>hello</servlet-name>
        <servlet-class>com.example.HelloServlet</servlet-class>
    </servlet>

    <servlet-mapping>
        <servlet-name>hello</servlet-name>
        <url-pattern>/</url-pattern>
    </servlet-mapping>

</web-app>
```

4. Jakarta Servlet Class (HelloServlet.java)

Create file at path: `src/main/java/com/example/HelloServlet.java`

```java
package com.example;

import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

public class HelloServlet extends HttpServlet {

    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse resp)
            throws ServletException, IOException {

        resp.setContentType("text/plain");

        PrintWriter out = resp.getWriter();
        out.println("Hello, Google App Engine with Jakarta Servlet 6!");
    }
}
```

5. Project Directory Structure

Ensure your standard Maven project layout matches the exact structure below:

```
gae-java-hello/
├── pom.xml
└── src/
    └── main/
        ├── java/
        │   └── com/
        │       └── example/
        │           └── HelloServlet.java        <-- Servlet Class
        └── webapp/
            └── WEB-INF/
                ├── appengine-web.xml        <-- GAE Runtime Config
                └── web.xml                  <-- Web Deployment Descriptor
```

6. Local Execution & Testing

Run the local Jetty server from your terminal inside the project root directory:

```
mvn clean jetty:run
```

> **Server Started Output:** Look for lines similar to `[INFO] Started ServerConnector@... {HTTP/1.1}{0.0.0.0:8080}`. Access your app at **http://localhost:8080**.

7. Deploying to Google Cloud App Engine

Step	Command / Instruction
1. Authenticate	`gcloud auth application-default login`
2. Set GCP Project	`gcloud config set project YOUR_PROJECT_ID`
3. Build Package	`mvn clean package`
4. Deploy App	`mvn appengine:deploy`

> **Deployment Note**
>
> The `appengine:deploy` goal automatically uploads your WAR package and deploys it to your configured Google Cloud App Engine standard environment using Java 17.